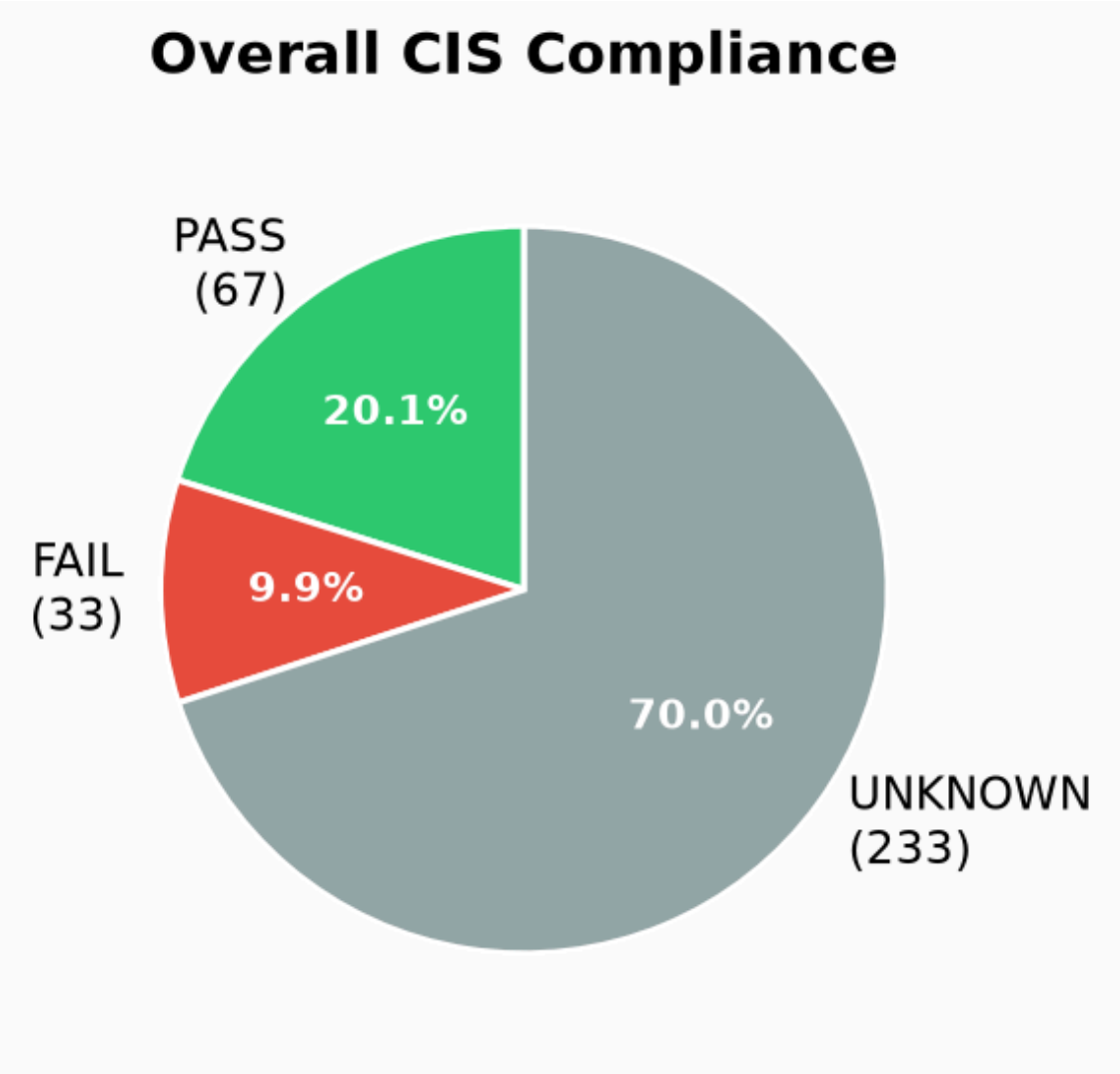


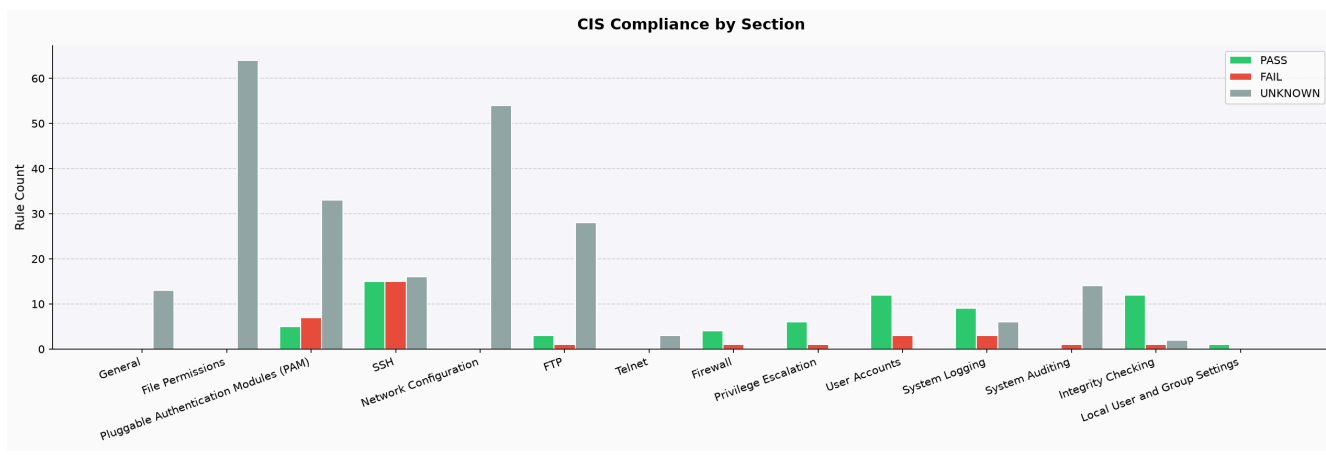
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	333
✓ Passed	67
✗ Failed	33
? Unknown	233
Overall Pass Rate	20.1%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
General	0	0	13	0%
File Permissions	0	0	64	0%
Pluggable Authentication Modules (PAM)	5	7	33	11%
SSH	15	15	16	33%
Network Configuration	0	0	54	0%
FTP	3	1	28	9%
Telnet	0	0	3	0%
Firewall	4	1	0	80%
Privilege Escalation	6	1	0	86%
User Accounts	12	3	0	80%
System Logging	9	3	6	50%
System Auditing	0	1	14	0%
Integrity Checking	12	1	2	80%
Local User and Group Settings	1	0	0	100%

CIS Benchmark Compliance Report

Here is the security compliance report based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and the machine's security profile.

Section 1.1 – Filesystem Kernel Modules & Partitions

Rule 1.1.1.1: Ensure cramfs kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.2: Ensure freevxfs kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.3: Ensure hfs kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.4: Ensure hfsplus kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.5: Ensure jffs2 kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.6: Ensure overlay kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.7: Ensure squashfs kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.8: Ensure udf kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing

error, preventing evaluation of kernel module status.

- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.9: Ensure firewire-core kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.10: Ensure usb-storage kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.1.11: Ensure unused filesystems kernel modules are not available (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing the collection of `lsmod_raw` data.
- **Recommendation:** Address the data collection error for the `filesystem` section. Once data is available, manually review the list of loaded kernel modules and disable any unused filesystem modules.

Rule 1.1.2.1.1: Ensure /tmp is tmpfs or a separate partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.1.2: Ensure nodev option set on /tmp partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.1.3: Ensure nosuid option set on /tmp partition (Automated)

- **Status:** UNKNOWN

- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.1.4: Ensure noexec option set on /tmp partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.2.1: Ensure /dev/shm is tmpfs or a separate partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.2.2: Ensure nodev option set on /dev/shm partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.2.3: Ensure nosuid option set on /dev/shm partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.2.4: Ensure noexec option set on /dev/shm partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.3.1: Ensure separate partition exists for /home (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.3.2: Ensure nodev option set on /home partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.3.3: Ensure nosuid option set on /home partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.4.1: Ensure separate partition exists for /var (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.4.2: Ensure nodev option set on /var partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.4.3: Ensure nosuid option set on /var partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.5.1: Ensure separate partition exists for /var/tmp (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to

gather necessary information for this rule.

Rule 1.1.2.5.2: Ensure nodev option set on /var/tmp partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.5.3: Ensure nosuid option set on /var/tmp partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.5.4: Ensure noexec option set on /var/tmp partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.6.1: Ensure separate partition exists for /var/log (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.6.2: Ensure nodev option set on /var/log partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.6.3: Ensure nosuid option set on /var/log partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.

- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.6.4: Ensure noexec option set on /var/log partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.7.1: Ensure separate partition exists for /var/log/audit (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.7.2: Ensure nodev option set on /var/log/audit partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.7.3: Ensure nosuid option set on /var/log/audit partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Rule 1.1.2.7.4: Ensure noexec option set on /var/log/audit partition (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile reported a JSON parsing error, preventing evaluation of partition configuration.
- **Recommendation:** Address the data collection error for the `filesystem` section to gather necessary information for this rule.

Section 1.2 – Package Management

Rule 1.2.1.1: Ensure the source.list and .source files use the Signed-By option (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing the collection of `sources_files` and `matching_signed_by_lines` data.
- **Recommendation:** Address the data collection error for the `package_management` section. Once data is available, manually verify that all `source.list` and `.source` files use the `Signed-By` option to ensure package authenticity.

Rule 1.2.1.2: Ensure weak dependencies are configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of weak dependencies configuration.
- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.1.3: Ensure access to gpg key files are configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of GPG key file access.
- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.1.4: Ensure access to /etc/apt/trusted.gpg.d directory is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/apt/trusted.gpg.d` directory access.
- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.1.5: Ensure access to /etc/apt/auth.conf.d directory is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/apt/auth.conf.d` directory access.
- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.1.6: Ensure access to files in the /etc/apt/auth.conf.d/ directory is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of files in `/etc/apt/auth.conf.d/`.

- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.1.7: Ensure access to `/usr/share/keyrings` directory is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of `/usr/share/keyrings` directory access.
- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.1.8: Ensure access to `/etc/apt/sources.list.d` directory is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/apt/sources.list.d` directory access.
- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.1.9: Ensure access to files in `/etc/apt/sources.list.d` are configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing evaluation of files in `/etc/apt/sources.list.d`.
- **Recommendation:** Address the data collection error for the `package_management` section to gather necessary information for this rule.

Rule 1.2.2.1: Ensure updates, patches, and additional security software are installed (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The `package_management` object in the security profile reported a JSON parsing error, preventing the collection of package update status.
- **Recommendation:** Address the data collection error for the `package_management` section. Once data is available, manually verify that all system updates, patches, and security software are installed and up-to-date.

Section 1.3 – Mandatory Access Control

Rule 1.3.1.1: Ensure apparmor packages are installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile reported a JSON parsing error, preventing evaluation of AppArmor package installation.
- **Recommendation:** Address the data collection error for the `apparmor` section to gather necessary information for this rule.

Rule 1.3.1.2: Ensure AppArmor is enabled (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile reported a JSON parsing error, preventing evaluation of AppArmor enablement.
- **Recommendation:** Address the data collection error for the `apparmor` section to gather necessary information for this rule.

Rule 1.3.1.3: Ensure all AppArmor Profiles are enforcing (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile reported a JSON parsing error, preventing evaluation of AppArmor profile status.
- **Recommendation:** Address the data collection error for the `apparmor` section to gather necessary information for this rule.

Rule 1.3.1.4: Ensure `apparmor_restrict_unprivileged_unconfined` is enabled (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile reported a JSON parsing error, preventing evaluation of `apparmor_restrict_unprivileged_unconfined` status.
- **Recommendation:** Address the data collection error for the `apparmor` section to gather necessary information for this rule.

Section 1.4 – Configure Bootloader

Rule 1.4.1: Ensure bootloader password is set (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `bootloader` object in the security profile reported a JSON parsing error, preventing evaluation of bootloader password status.
- **Recommendation:** Address the data collection error for the `bootloader` section to gather necessary information for this rule.

Rule 1.4.2: Ensure access to bootloader config is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `bootloader` object in the security profile reported a JSON parsing error, preventing evaluation of bootloader configuration file access.
- **Recommendation:** Address the data collection error for the `bootloader` section to gather necessary information for this rule.

Section 1.5 – Configure Additional Process Hardening

Rule 1.5.1: Ensure `fs.protected_hardlinks` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `fs.protected_hardlinks` status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.2: Ensure fs.protected_symlinks is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `fs.protected_symlinks` status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.3: Ensure kernel.yama.ptrace_scope is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `kernel.yama.ptrace_scope` status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.4: Ensure fs.suid_dumpable is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `fs.suid_dumpable` status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.5: Ensure kernel.dmesg_restrict is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `kernel.dmesg_restrict` status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.6: Ensure prelink is not installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `prelink` installation status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.7: Ensure Automatic Error Reporting is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of Automatic Error Reporting status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.8: Ensure kernel.kptr_restrict is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON

parsing error, preventing evaluation of `kernel.kptr_restrict` status.

- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.9: Ensure `kernel.randomize_va_space` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `kernel.randomize_va_space` status.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.10: Ensure core file size is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of core file size configuration.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.11: Ensure `systemd-coredump ProcessSizeMax` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `systemd-coredump ProcessSizeMax` configuration.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Rule 1.5.12: Ensure `systemd-coredump Storage` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile reported a JSON parsing error, preventing evaluation of `systemd-coredump Storage` configuration.
- **Recommendation:** Address the data collection error for the `process_hardening` section to gather necessary information for this rule.

Section 1.6 – Configure Command Line Warning Banners

Rule 1.6.1: Ensure `/etc/motd` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/motd` configuration.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.2: Ensure `/etc/issue` is configured (Automated)

- **Status:** UNKNOWN

- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/issue` configuration.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.3: Ensure `/etc/issue.net` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/issue.net` configuration.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.4: Ensure `pam_motd` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `pam_motd` configuration.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.5: Ensure `sshd` warning Banner is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is "none".
- **Recommendation:** Configure a warning banner by setting `Banner /etc/issue.net` in `/etc/ssh/sshd_config` and ensure `/etc/issue.net` contains an appropriate warning message.

Rule 1.6.6: Ensure access to `/etc/motd` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/motd` access.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.7: Ensure access to `/etc/issue` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/issue` access.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.8: Ensure access to `/etc/issue.net` is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/issue.net` access.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.9: Ensure access to pam_motd file is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of `pam_motd` file access.
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Rule 1.6.10: Ensure access to sshd warning banner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile reported a JSON parsing error, preventing evaluation of SSH banner file access. (Note: SSH banner content is covered in 5.1.5).
- **Recommendation:** Address the data collection error for the `warning_banners` section to gather necessary information for this rule.

Section 1.7 – Configure GNOME Display Manager

Rule 1.7.1: Ensure GDM login banner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported a JSON parsing error, preventing evaluation of GDM login banner configuration.
- **Recommendation:** Address the data collection error for the `gnome` section to gather necessary information for this rule.

Rule 1.7.2: Ensure GDM disable-user-list is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported a JSON parsing error, preventing evaluation of GDM `disable-user-list` configuration.
- **Recommendation:** Address the data collection error for the `gnome` section to gather necessary information for this rule.

Rule 1.7.3: Ensure GDM screen lock is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported a JSON parsing error, preventing evaluation of GDM screen lock configuration.
- **Recommendation:** Address the data collection error for the `gnome` section to gather necessary information for this rule.

Rule 1.7.4: Ensure GDM automount is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported a JSON parsing error, preventing evaluation of GDM automount configuration.
- **Recommendation:** Address the data collection error for the `gnome` section to gather necessary information for this rule.

Rule 1.7.5: Ensure GDM autorun-never is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported a JSON parsing error, preventing evaluation of GDM `autorun-never` configuration.
- **Recommendation:** Address the data collection error for the `gnome` section to gather necessary information for this rule.

Rule 1.7.6: Ensure XDMCP is not enabled (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported a JSON parsing error, preventing evaluation of XDMCP status.
- **Recommendation:** Address the data collection error for the `gnome` section to gather necessary information for this rule.

Rule 1.7.7: Ensure Xwayland is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile reported a JSON parsing error, preventing evaluation of Xwayland configuration.
- **Recommendation:** Address the data collection error for the `gnome` section to gather necessary information for this rule.

Section 2 – Services

Rule 2.1.1: Ensure autofs services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of autofs services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.2: Ensure mail transfer agents are configured for local-only mode (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of MTA configuration.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.3: Ensure avahi daemon services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of avahi daemon services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.4: Ensure only approved services are listening on a network interface (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing the collection of listening sockets data.
- **Recommendation:** Address the data collection error for the `services` section. Once data is available, manually review the list of listening services and ensure only approved services are active.

Rule 2.1.5: Ensure dhcp server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of DHCP server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.6: Ensure web server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of web server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.7: Ensure dns server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of DNS server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.8: Ensure ftp server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of FTP server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.9: Ensure dnsmasq services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of dnsmasq services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.10: Ensure ldap server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of LDAP server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.11: Ensure message access server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of message access server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.12: Ensure network file system services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of NFS services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.13: Ensure nis server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of NIS server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.14: Ensure print server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of print server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.15: Ensure rpcbind services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of rpcbind services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.16: Ensure rsync services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error,

preventing evaluation of rsync services.

- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.17: Ensure samba file server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of Samba services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.18: Ensure snmp services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of SNMP services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.19: Ensure telnet server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of Telnet server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.20: Ensure tftp server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of TFTP server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.21: Ensure web proxy server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of web proxy server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.22: Ensure xinetd services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of xinetd services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.1.23: Ensure X window server services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of X window server services.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.2.1: Ensure nis client is not installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of NIS client installation.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.2.2: Ensure rsh client is not installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of RSH client installation.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.2.3: Ensure talk client is not installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of talk client installation.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.2.4: Ensure telnet client is not installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of Telnet client installation.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.2.5: Ensure ldap client is not installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error, preventing evaluation of LDAP client installation.
- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.2.6: Ensure ftp client is not installed (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile reported a JSON parsing error,

preventing evaluation of FTP client installation.

- **Recommendation:** Address the data collection error for the `services` section to gather necessary information for this rule.

Rule 2.3.1.1: Ensure a single time synchronization daemon is in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported a JSON parsing error, preventing evaluation of time synchronization daemon status.
- **Recommendation:** Address the data collection error for the `time_sync` section to gather necessary information for this rule.

Rule 2.3.2.1: Ensure systemd-timesyncd configured with authorized timeserver (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported a JSON parsing error, preventing evaluation of `systemd-timesyncd` configuration.
- **Recommendation:** Address the data collection error for the `time_sync` section to gather necessary information for this rule.

Rule 2.3.2.2: Ensure systemd-timesyncd is enabled and running (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported a JSON parsing error, preventing evaluation of `systemd-timesyncd` service status.
- **Recommendation:** Address the data collection error for the `time_sync` section to gather necessary information for this rule.

Rule 2.3.3.1: Ensure chrony is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported a JSON parsing error, preventing evaluation of chrony configuration.
- **Recommendation:** Address the data collection error for the `time_sync` section to gather necessary information for this rule.

Rule 2.3.3.2: Ensure chrony is running as user _chrony (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported a JSON parsing error, preventing evaluation of chrony process user.
- **Recommendation:** Address the data collection error for the `time_sync` section to gather necessary information for this rule.

Rule 2.3.3.3: Ensure chrony is enabled and running (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile reported a JSON parsing error,

preventing evaluation of chrony service status.

- **Recommendation:** Address the data collection error for the `time_sync` section to gather necessary information for this rule.

Rule 2.4.1.1: Ensure cron daemon is enabled and active (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of cron daemon status.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.2: Ensure access to /etc/crontab is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/crontab` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.3: Ensure access to /etc/cron.hourly is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/cron.hourly` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.4: Ensure access to /etc/cron.daily is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/cron.daily` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.5: Ensure access to /etc/cron.weekly is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/cron.weekly` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.6: Ensure access to /etc/cron.monthly is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/cron.monthly` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.7: Ensure access to /etc/cron.yearly is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/cron.yearly` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.8: Ensure access to /etc/cron.d is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `/etc/cron.d` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.1.9: Ensure access to crontab is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `crontab` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Rule 2.4.2.1: Ensure access to at is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile reported a JSON parsing error, preventing evaluation of `at` access.
- **Recommendation:** Address the data collection error for the `job_schedulers` section to gather necessary information for this rule.

Section 3 – Network Configuration

Rule 3.1.1: Ensure IPv6 status is identified (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing the collection of IPv6 status data.
- **Recommendation:** Address the data collection error for the `network_config` section. Once data is available, manually review the IPv6 status to determine if it is enabled or disabled according to policy.

Rule 3.1.2: Ensure wireless interfaces are not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of wireless interfaces.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.1.3: Ensure bluetooth services are not in use (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of Bluetooth services.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.2.1: Ensure atm kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.2.2: Ensure can kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.2.3: Ensure dccp kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.2.4: Ensure rds kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.2.5: Ensure sctp kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of kernel module status.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.2.6: Ensure tipc kernel module is not available (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON

parsing error, preventing evaluation of kernel module status.

- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.1: Ensure net.ipv4.ip_forward is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.2: Ensure net.ipv4.conf.all.forwarding is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.3: Ensure net.ipv4.conf.default.forwarding is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.4: Ensure net.ipv4.conf.all.send_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.5: Ensure net.ipv4.conf.default.send_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.6: Ensure net.ipv4.icmp_ignore_bogus_error_responses is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.7: Ensure net.ipv4.icmp_echo_ignore_broadcasts is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.8: Ensure net.ipv4.conf.all.accept_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.9: Ensure net.ipv4.conf.default.accept_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.10: Ensure net.ipv4.conf.all.secure_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.11: Ensure net.ipv4.conf.default.secure_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.12: Ensure net.ipv4.conf.all.rp_filter is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.13: Ensure net.ipv4.conf.default.rp_filter is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.14: Ensure net.ipv4.conf.all.accept_source_route is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.15: Ensure net.ipv4.conf.default.accept_source_route is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.16: Ensure net.ipv4.conf.all.log_martians is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.1.17: Ensure net.ipv4.conf.default.log_martians is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

to gather necessary information for this rule.

Rule 3.3.1.18: Ensure net.ipv4.tcp_syncookies is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv4 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.1: Ensure net.ipv6.conf.all.forwarding is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv6 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.2: Ensure net.ipv6.conf.default.forwarding is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv6 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.3: Ensure net.ipv6.conf.all.accept_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv6 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.4: Ensure net.ipv6.conf.default.accept_redirects is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv6 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.5: Ensure net.ipv6.conf.all.accept_source_route is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON

parsing error, preventing evaluation of IPv6 kernel parameters.

- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.6: Ensure net.ipv6.conf.default.accept_source_route is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv6 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.7: Ensure net.ipv6.conf.all.accept_ra is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv6 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Rule 3.3.2.8: Ensure net.ipv6.conf.default.accept_ra is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile reported a JSON parsing error, preventing evaluation of IPv6 kernel parameters.
- **Recommendation:** Address the data collection error for the `network_config` section to gather necessary information for this rule.

Section 4.1 – Configure Uncomplicated Firewall

Rule 4.1.1: Ensure ufw is installed (Automated)

- **Status:** PASS
- **Evidence:** `ufw_firewall.ufw_installed` is `true`, and `ufw_firewall.ufw_dpkg_status` is `"install ok installed"`.
- **Recommendation:** None.

Rule 4.1.2: Ensure ufw service is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ufw_firewall.service_configured.is_enabled` is `"enabled"` and `ufw_firewall.service_configured.is_active` is `"active"`, but `ufw_firewall.service_configured.status_verbose` is `"Status: inactive"`.
- **Recommendation:** Investigate why the UFW service reports as `"inactive"` despite being enabled and active. Ensure UFW is properly started and enforcing rules. Run `sudo ufw enable` and `sudo ufw status verbose` to confirm.

Rule 4.1.3: Ensure ufw incoming default is configured (Automated)

- **Status:** PASS
- **Evidence:**
`ufw_firewall.default_policies.etc_default_ufw.DEFAULT_INPUT_POLICY`
is "DROP".
- **Recommendation:** None.

Rule 4.1.4: Ensure ufw outgoing default is configured (Automated)

- **Status:** PASS
- **Evidence:**
`ufw_firewall.default_policies.etc_default_ufw.DEFAULT_OUTPUT_POLICY`
is "ACCEPT".
- **Recommendation:** None.

Rule 4.1.5: Ensure ufw routed default is configured (Automated)

- **Status:** PASS
- **Evidence:**
`ufw_firewall.default_policies.etc_default_ufw.DEFAULT_FORWARD_POLICY`
is "DROP".
- **Recommendation:** None.

Section 5.1 – Configure SSH Server

Rule 5.1.1: Ensure access to /etc/ssh/sshd_config is configured (Automated)

- **Status:** PASS
- **Evidence:** File `/etc/ssh/sshd_config` has permissions 0644, owner root, group root.
- **Recommendation:** None.

Rule 5.1.2: Ensure access to SSH private host key files is configured (Automated)

- **Status:** PASS
- **Evidence:** All SSH private host key files have permissions 0600, owner root, group root.
- **Recommendation:** None.

Rule 5.1.3: Ensure access to SSH public host key files is configured (Automated)

- **Status:** PASS
- **Evidence:** All SSH public host key files have permissions 0644, owner root, group root.
- **Recommendation:** None.

Rule 5.1.4: Ensure sshd access is configured (Automated)

- **Status:** FAIL

- **Evidence:** `ssh.authentication.password_authentication` is true (risk: high). `ssh.authentication.permit_root_login` is "without-password".
- **Recommendation:** Disable password authentication by setting `PasswordAuthentication no` in `/etc/ssh/sshd_config`. Ensure `PermitRootLogin no` is set. Use key-based authentication instead.

Rule 5.1.5: Ensure sshd Banner is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is "none".
- **Recommendation:** Configure a warning banner by setting `Banner /etc/issue.net` in `/etc/ssh/sshd_config` and ensure `/etc/issue.net` contains an appropriate warning message.

Rule 5.1.6: Ensure sshd Ciphers are configured (Automated)

- **Status:** PASS
- **Evidence:** Configured ciphers are `chacha20-poly1305@openssh.com`, `aes128-ctr`, `aes192-ctr`, `aes256-ctr`, `aes128-gcm@openssh.com`, `aes256-gcm@openssh.com`. No weak ciphers detected.
- **Recommendation:** None.

Rule 5.1.7: Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured (Automated)

- **Status:** FAIL
- **Evidence:** `ClientAliveInterval` is 0. `ClientAliveCountMax` is 3.
- **Recommendation:** Set `ClientAliveInterval` to a reasonable value (e.g., 300 seconds) in `/etc/ssh/sshd_config` to automatically terminate idle SSH sessions.

Rule 5.1.8: Ensure sshd DisableForwarding is enabled (Automated)

- **Status:** FAIL
- **Evidence:** `ssh.forwarding.allow_tcp_forwarding` is true, `ssh.forwarding.allow_agent_forwarding` is true, `ssh.forwarding.allow_streamlocal_forwarding` is true.
- **Recommendation:** Disable all types of forwarding by setting `AllowTcpForwarding no`, `AllowAgentForwarding no`, and `AllowStreamLocalForwarding no` in `/etc/ssh/sshd_config`.

Rule 5.1.9: Ensure sshd GSSAPIAuthentication is disabled (Automated)

- **Status:** PASS
- **Evidence:** `GSSAPIAuthentication` is false.
- **Recommendation:** None.

Rule 5.1.10: Ensure sshd HostbasedAuthentication is disabled (Automated)

- **Status:** PASS
- **Evidence:** `HostbasedAuthentication` is false.
- **Recommendation:** None.

Rule 5.1.11: Ensure sshd IgnoreRhosts is enabled (Automated)

- **Status:** PASS
- **Evidence:** IgnoreRhosts is true.
- **Recommendation:** None.

Rule 5.1.12: Ensure sshd KexAlgorithms is configured (Automated)

- **Status:** FAIL
- **Evidence:** Weak KexAlgorithms detected: ecdh-sha2-nistp256, ecdh-sha2-nistp384, ecdh-sha2-nistp521 (high severity due to NIST/NSA backdoor concerns), diffie-hellman-group14-sha256 (medium severity, deprecated).
- **Recommendation:** Configure KexAlgorithms in /etc/ssh/sshd_config to exclude weak algorithms. Prioritize strong, modern key exchange algorithms (e.g., curve25519-sha256, diffie-hellman-group16-sha512). Refer to the CIS benchmark for a recommended list.

Rule 5.1.13: Ensure sshd LoginGraceTime is configured (Automated)

- **Status:** FAIL
- **Evidence:** LoginGraceTime is 120 seconds.
- **Recommendation:** Set LoginGraceTime to 60 seconds or less in /etc/ssh/sshd_config to reduce the window for brute-force attacks.

Rule 5.1.14: Ensure sshd LogLevel is configured (Automated)

- **Status:** PASS
- **Evidence:** LogLevel is "INFO".
- **Recommendation:** None.

Rule 5.1.15: Ensure sshd MACs are configured (Automated)

- **Status:** FAIL
- **Evidence:** Weak MACs detected: umac-64-etm@openssh.com, hmac-sha1-etm@openssh.com (high severity, Terrapin attack), umac-64@openssh.com, umac-128@openssh.com, hmac-sha2-256, hmac-sha2-512, hmac-sha1 (high severity, collision attacks).
- **Recommendation:** Configure MACs in /etc/ssh/sshd_config to exclude weak algorithms. Prioritize strong, modern MACs (e.g., hmac-sha2-512-etm@openssh.com, hmac-sha2-256-etm@openssh.com). Refer to the CIS benchmark for a recommended list and ensure mitigation for CVE-2023-48795 (Terrapin Attack).

Rule 5.1.16: Ensure sshd MaxAuthTries is configured (Automated)

- **Status:** FAIL
- **Evidence:** MaxAuthTries is 6.
- **Recommendation:** Set MaxAuthTries to 4 or less in /etc/ssh/sshd_config to limit the number of authentication attempts.

Rule 5.1.17: Ensure sshd MaxStartups is configured (Automated)

- **Status:** FAIL
- **Evidence:** MaxStartups is "10:30:100".
- **Recommendation:** Set MaxStartups to 10:30:60 or a similar value with a lower full threshold in /etc/ssh/sshd_config to prevent resource exhaustion from unauthenticated connections.

Rule 5.1.18: Ensure sshd MaxSessions is configured (Automated)

- **Status:** FAIL
- **Evidence:** MaxSessions is 10.
- **Recommendation:** Set MaxSessions to 4 or less in /etc/ssh/sshd_config to limit the number of open sessions per network connection.

Rule 5.1.19: Ensure sshd PermitEmptyPasswords is disabled (Automated)

- **Status:** PASS
- **Evidence:** PermitEmptyPasswords is false.
- **Recommendation:** None.

Rule 5.1.20: Ensure sshd PermitRootLogin is disabled (Automated)

- **Status:** FAIL
- **Evidence:** PermitRootLogin is "without-password".
- **Recommendation:** Set PermitRootLogin no in /etc/ssh/sshd_config. If root access is required, use sudo from a regular user account.

Rule 5.1.21: Ensure sshd PermitUserEnvironment is disabled (Automated)

- **Status:** PASS
- **Evidence:** PermitUserEnvironment is false.
- **Recommendation:** None.

Rule 5.1.22: Ensure sshd UsePAM is enabled (Automated)

- **Status:** PASS
- **Evidence:** UsePAM is true.
- **Recommendation:** None.

Rule 5.1.23: Ensure sshd post-quantum cryptography key exchange algorithms are configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The security profile does not contain specific information about post-quantum cryptography key exchange algorithms.
- **Recommendation:** Manually verify that post-quantum cryptography key exchange algorithms are configured in /etc/ssh/sshd_config if required by policy.

Rule 5.1.24: Ensure sshd ListenAddress is configured (Automated)

- **Status:** FAIL
- **Evidence:** ListenAddress is "0.0.0.0:22".
- **Recommendation:** Configure ListenAddress in /etc/ssh/sshd_config to bind

to specific network interfaces (e.g., `ListenAddress 192.168.1.10:22`) rather than all interfaces, if possible, to reduce exposure.

Section 5.2 – Configure Privilege Escalation

Rule 5.2.1: Ensure sudo is installed (Automated)

- **Status:** PASS
- **Evidence:** `sudo_installed.installed` is true, `version_output` is "Sudo version 1.9.15p5".
- **Recommendation:** None.

Rule 5.2.2: Ensure sudo commands use pty (Automated)

- **Status:** PASS
- **Evidence:** `/etc/sudoers` contains "Defaults use_pty".
- **Recommendation:** None.

Rule 5.2.3: Ensure sudo log file exists (Automated)

- **Status:** PASS
- **Evidence:** Sudo logging is configured via syslog/journald, with `auth.log` and `syslog` existing, `rsyslog_authpriv_configured` is true, and `journald_sudo_evidence` shows sudo entries.
- **Recommendation:** None.

Rule 5.2.4: Ensure users must provide password for escalation (Automated)

- **Status:** PASS
- **Evidence:** No `NOPASSWD` or `NOAUTHENTICATE` entries found in sudoers files.
- **Recommendation:** None.

Rule 5.2.5: Ensure re-authentication for privilege escalation is not disabled globally (Automated)

- **Status:** PASS
- **Evidence:** No `NOAUTHENTICATE` entries found in sudoers files.
- **Recommendation:** None.

Rule 5.2.6: Ensure sudo timestamp_timeout is configured (Automated)

- **Status:** PASS
- **Evidence:** No `timestamp_timeout` entries found in sudoers files, implying the default (5 minutes) is in effect.
- **Recommendation:** None.

Rule 5.2.7: Ensure access to the su command is restricted (Automated)

- **Status:** FAIL
- **Evidence:** The `pam_wheel.so` module is commented out in `/etc/pam.d/su`,

meaning `su` access is not restricted to a specific group.

- **Recommendation:** Uncomment the `auth required pam_wheel.so` line in `/etc/pam.d/su` and ensure that only authorized users are members of the `wheel` group.

Section 5.3 – Pluggable Authentication Modules (PAM)

Rule 5.3.1.1: Ensure latest version of pam is installed (Automated)

- **Status:** PASS
- **Evidence:** `libpam-runtime` is installed (`1.5.3-5ubuntu5.5`) and matches the candidate version.
- **Recommendation:** None.

Rule 5.3.1.2: Ensure latest version of libpam-modules is installed (Automated)

- **Status:** PASS
- **Evidence:** `libpam-modules` is installed (`1.5.3-5ubuntu5.5`) and matches the candidate version.
- **Recommendation:** None.

Rule 5.3.1.3: Ensure latest version of libpam-pwquality is installed (Automated)

- **Status:** FAIL
- **Evidence:** `libpam-pwquality` is not installed.
- **Recommendation:** Install the `libpam-pwquality` package using `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: Ensure latest version of cracklib-runtime is installed (Automated)

- **Status:** FAIL
- **Evidence:** `cracklib-runtime` is not installed.
- **Recommendation:** Install the `cracklib-runtime` package using `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: Ensure pam_unix module is enabled (Automated)

- **Status:** PASS
- **Evidence:** `pam_unix.so` module is enabled with multiple active lines in `common-auth`, `common-account`, `common-password`, and `common-session`.
- **Recommendation:** None.

Rule 5.3.2.2: Ensure pam_faillock module is enabled (Automated)

- **Status:** FAIL
- **Evidence:** `pam_faillock.so` module is not enabled; no active lines found.
- **Recommendation:** Enable `pam_faillock.so` in the PAM configuration (e.g., via `pam-auth-update` or by manually adding lines to `common-auth` and `common-account`).

Rule 5.3.2.3: Ensure pam_pwquality module is enabled (Automated)

- **Status:** FAIL
- **Evidence:** `pam_pwquality.so` module is not enabled; no active lines found.
- **Recommendation:** Enable `pam_pwquality.so` in the PAM configuration (e.g., via `pam-auth-update` or by manually adding lines to `common-password`). This also requires `libpam-pwquality` to be installed (see 5.3.1.3).

Rule 5.3.2.4: Ensure pam_pwhistory module is enabled (Automated)

- **Status:** FAIL
- **Evidence:** `pam_pwhistory.so` module is not enabled; no active lines found.
- **Recommendation:** Enable `pam_pwhistory.so` in the PAM configuration (e.g., via `pam-auth-update` or by manually adding lines to `common-password`).

Rule 5.3.3.1.1: Ensure password failed attempts lockout is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_faillock` module is not enabled, and no `deny` setting is found in `faillock.conf` or inline PAM settings.
- **Recommendation:** Enable `pam_faillock` and configure `deny = 5` or less in `/etc/security/faillock.conf` or in the PAM configuration files.

Rule 5.3.3.1.2: Ensure password unlock time is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_faillock` module is not enabled, and no `unlock_time` setting is found in `faillock.conf` or inline PAM settings.
- **Recommendation:** Enable `pam_faillock` and configure `unlock_time = 900` or `0` in `/etc/security/faillock.conf` or in the PAM configuration files.

Rule 5.3.3.1.3: Ensure password failed attempts lockout includes root account (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_faillock` module is not enabled, and no `even_deny_root` or `root_unlock_time` setting is found.
- **Recommendation:** Enable `pam_faillock` and configure `even_deny_root` and `root_unlock_time = 60` or greater in `/etc/security/faillock.conf` or in the PAM configuration files.

Rule 5.3.3.2.1: Ensure password number of changed characters is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwquality` module is not enabled, and no `difok` setting is found.
- **Recommendation:** Enable `pam_pwquality` and configure `difok = 2` or greater in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.2.2: Ensure password length is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwquality` module is not enabled, and no `minlen` setting is found.
- **Recommendation:** Enable `pam_pwquality` and configure `minlen = 14` or greater in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.2.3: Ensure password complexity is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** `pam_pwquality` module is not enabled, and no `dcredit`, `ucredit`, `lcredit`, `ocredit` settings are found.
- **Recommendation:** Enable `pam_pwquality` and configure password complexity requirements (e.g., `dcredit=-1`, `ucredit=-1`, `lcredit=-1`, `ocredit=-1` for one of each type) in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.2.4: Ensure password same consecutive characters is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwquality` module is not enabled, and no `maxrepeat` setting is found.
- **Recommendation:** Enable `pam_pwquality` and configure `maxrepeat = 3` or less (or 0 to disable) in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.2.5: Ensure password maximum sequential characters is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwquality` module is not enabled, and no `maxsequence` setting is found.
- **Recommendation:** Enable `pam_pwquality` and configure `maxsequence = 3` or less (or 0 to disable) in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.2.6: Ensure password dictionary check is enabled (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwquality` module is not enabled, and no `dictcheck` setting is found.
- **Recommendation:** Enable `pam_pwquality` and configure `dictcheck = 1` in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.2.7: Ensure password quality checking is enforced (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwquality` module is not enabled, and no `enforcing` setting is found.
- **Recommendation:** Enable `pam_pwquality` and configure `enforcing = 1` in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.2.8: Ensure password quality is enforced for the root user

(Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwquality` module is not enabled, and no `enforce_for_root` setting is found.
- **Recommendation:** Enable `pam_pwquality` and configure `enforce_for_root` in `/etc/security/pwquality.conf` or in the PAM configuration files.

Rule 5.3.3.3.1: Ensure password history remember is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwhistory` module is not enabled, and no `remember` setting is found.
- **Recommendation:** Enable `pam_pwhistory` and configure `remember = 24` or greater in the PAM configuration files (e.g., `common-password`).

Rule 5.3.3.3.2: Ensure password history is enforced for the root user (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwhistory` module is not enabled, and no `enforce_for_root` setting is found.
- **Recommendation:** Enable `pam_pwhistory` and include the `enforce_for_root` option in the `pam_pwhistory` line in the PAM configuration files.

Rule 5.3.3.3.3: Ensure pam_pwhistory includes use_authtok (Automated)

- **Status:** UNKNOWN
- **Evidence:** `pam_pwhistory` module is not enabled, and no `use_authtok` setting is found.
- **Recommendation:** Enable `pam_pwhistory` and include the `use_authtok` option in the `pam_pwhistory` line in the PAM configuration files.

Rule 5.3.3.4.1: Ensure pam_unix does not include nullok (Automated)

- **Status:** FAIL
- **Evidence:** The `nullok` option is present in the `pam_unix.so` line in `/etc/pam.d/common-auth`.
- **Recommendation:** Remove the `nullok` option from the `pam_unix.so` line in `/etc/pam.d/common-auth`.

Rule 5.3.3.4.2: Ensure pam_unix does not include remember (Automated)

- **Status:** PASS
- **Evidence:** The `remember` option is not present in any `pam_unix.so` line.
- **Recommendation:** None.

Rule 5.3.3.4.3: Ensure pam_unix includes a strong password hashing algorithm (Automated)

- **Status:** PASS

- **Evidence:** The `yescrypt` hashing algorithm is present in the `pam_unix.so` line in `/etc/pam.d/common-password`.
- **Recommendation:** None.

Rule 5.3.3.4.4: Ensure pam_unix includes use_authtok (Automated)

- **Status:** FAIL
- **Evidence:** The `use_authtok` option is not present in the `pam_unix.so` line in `/etc/pam.d/common-password`.
- **Recommendation:** Add the `use_authtok` option to the `pam_unix.so` line in `/etc/pam.d/common-password` to ensure that the new password is used for all modules in the stack.

Section 5.4 – User Accounts and Environment

Rule 5.4.1.1: Ensure password expiration is configured (Automated)

- **Status:** FAIL
- **Evidence:** `PASS_MAX_DAYS` is set to 99999 in `/etc/login.defs`, indicating passwords do not expire.
- **Recommendation:** Set `PASS_MAX_DAYS` to 90 or less in `/etc/login.defs` to enforce regular password changes.

Rule 5.4.1.2: Ensure minimum password days is configured (Manual)

- **Status:** FAIL
- **Evidence:** `PASS_MIN_DAYS` is set to 0 in `/etc/login.defs`, allowing immediate password changes.
- **Recommendation:** Set `PASS_MIN_DAYS` to 7 or more in `/etc/login.defs` to prevent users from cycling through a small set of passwords.

Rule 5.4.1.3: Ensure password expiration warning days is configured (Automated)

- **Status:** PASS
- **Evidence:** `PASS_WARN_AGE` is set to 7 in `/etc/login.defs`.
- **Recommendation:** None.

Rule 5.4.1.4: Ensure strong password hashing algorithm is configured (Automated)

- **Status:** PASS
- **Evidence:** `ENCRYPT_METHOD` is set to "SHA512" in `/etc/login.defs`.
- **Recommendation:** None.

Rule 5.4.1.5: Ensure inactive password lock is configured (Automated)

- **Status:** FAIL
- **Evidence:** `INACTIVE` is set to -1 in `/etc/default/useradd` (implied by `useradd_inactive`), meaning accounts never become inactive.
- **Recommendation:** Set `INACTIVE` to 30 or less in `/etc/default/useradd` to lock

accounts after a period of inactivity.

Rule 5.4.1.6: Ensure all users last password change date is in the past (Automated)

- **Status:** PASS
- **Evidence:** All user accounts, including `root`, `sebti`, `sshd`, and `ftp`, have their `last_password_change` date in the past.
- **Recommendation:** None.

Rule 5.4.2.1: Ensure root is the only UID 0 account (Automated)

- **Status:** PASS
- **Evidence:** Only `root` is listed as a UID 0 account.
- **Recommendation:** None.

Rule 5.4.2.2: Ensure root is the only GID 0 account (Automated)

- **Status:** PASS
- **Evidence:** Only `root` is listed as a primary GID 0 account.
- **Recommendation:** None.

Rule 5.4.2.3: Ensure group root is the only GID 0 group (Automated)

- **Status:** PASS
- **Evidence:** Only `root` is listed as a GID 0 group.
- **Recommendation:** None.

Rule 5.4.2.4: Ensure root account access is controlled (Automated)

- **Status:** FAIL
- **Evidence:** `PermitRootLogin` is "without-password" in SSH, and `/etc/securetty` does not exist, which means direct root login is not restricted to specific TTYs.
- **Recommendation:** Set `PermitRootLogin no` in `/etc/ssh/sshd_config`. Create and configure `/etc/securetty` to restrict direct root login to only secure TTYs, if direct console login is necessary.

Rule 5.4.2.5: Ensure root path integrity (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** Root's PATH is `/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin`.
- **Recommendation:** Manually verify that all directories in root's PATH are owned by root and writable only by root. Remove any unnecessary directories from root's PATH, especially those that might be user-writable or contain untrusted executables, such as `/snap/bin` if not strictly necessary for root operations.

Rule 5.4.2.6: Ensure root user umask is configured (Automated)

- **Status:** PASS
- **Evidence:** `UMASK` is set to `022` in `/etc/login.defs`.
- **Recommendation:** None.

Rule 5.4.2.7: Ensure system accounts do not have a valid login shell (Automated)

- **Status:** FAIL
- **Evidence:** The `sync` system account has `/bin/sync` as its login shell, which is a valid shell.
- **Recommendation:** Change the login shell for the `sync` account to `/usr/sbin/nologin` or `/bin/false` using `usermod -s /usr/sbin/nologin sync`.

Rule 5.4.2.8: Ensure accounts without a valid login shell are locked (Automated)

- **Status:** PASS
- **Evidence:** All listed system accounts (e.g., `daemon`, `bin`, `sys`, `games`, `mail`, `www-data`) have a `passwd_status` of "L", indicating they are locked.
- **Recommendation:** None.

Rule 5.4.3.1: Ensure nologin is not listed in /etc/shells (Automated)

- **Status:** PASS
- **Evidence:** `/usr/sbin/nologin` is not listed in `/etc/shells`.
- **Recommendation:** None.

Rule 5.4.3.2: Ensure default user shell timeout is configured (Automated)

- **Status:** FAIL
- **Evidence:** No global `TMOUT` setting found in `/etc/profile`, `/etc/profile.d/`, or `/etc/bash.bashrc`.
- **Recommendation:** Set `TMOUT=900` (or less) in `/etc/profile.d/timeout.sh` to automatically log out idle shell sessions.

Rule 5.4.3.3: Ensure default user umask is configured (Automated)

- **Status:** PASS
- **Evidence:** `UMASK` is set to `022` in `/etc/login.defs`, and `pam_umask.so` is configured in `common-session`.
- **Recommendation:** None.

Section 6.1 – System Logging

Rule 6.1.1.1.1: Ensure journald service is active (Automated)

- **Status:** PASS
- **Evidence:** `systemd-journald` service is active.
- **Recommendation:** None.

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use (Automated)

- **Status:** PASS
- **Evidence:** `systemd-journal-remote` is not installed, and `systemd-journal-`

upload and systemd-journal-remote.socket are inactive/not found.

- **Recommendation:** None.

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog (Automated)

- **Status:** PASS
- **Evidence:** ForwardToSyslog is commented out in /etc/systemd/journald.conf, implying the default of yes is in effect.
- **Recommendation:** None.

Rule 6.1.1.1.4: Ensure journald log file access is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** /run/log/journal and /var/log/journal directories have mode 2755, owner root, group systemd-journal (GID 999).
- **Recommendation:** Manually review and ensure that access to journald log files and directories is restricted to authorized users. Consider changing directory permissions to 0750 if appropriate for your environment, and ensure only root and adm (or systemd-journal) have access.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** Journald log rotation settings (SystemMaxUse, SystemMaxFiles, MaxRetentionSec, MaxFileSec) are commented out in /etc/systemd/journald.conf, relying on default values.
- **Recommendation:** Manually review and configure journald log rotation settings in /etc/systemd/journald.conf (e.g., SystemMaxUse, MaxRetentionSec) to meet organizational retention policies.

Rule 6.1.1.1.6: Ensure journald Storage is configured (Automated)

- **Status:** PASS
- **Evidence:** Storage is commented out in /etc/systemd/journald.conf, defaulting to auto. Since /var/log/journal exists, logs are stored persistently.
- **Recommendation:** None. (Optionally, explicitly set Storage=persistent for clarity).

Rule 6.1.1.1.7: Ensure journald Compress is configured (Automated)

- **Status:** PASS
- **Evidence:** Compress is commented out in /etc/systemd/journald.conf, implying the default of yes (compression enabled) is in effect.
- **Recommendation:** None.

Rule 6.1.2.1: Ensure rsyslog is installed (Automated)

- **Status:** PASS
- **Evidence:** rsyslog_installed is true.
- **Recommendation:** None.

Rule 6.1.2.2: Ensure rsyslog service is enabled and active (Automated)

- **Status:** PASS
- **Evidence:** rsyslog service is enabled and active.
- **Recommendation:** None.

Rule 6.1.2.3: Ensure rsyslog log file creation mode is configured (Automated)

- **Status:** PASS
- **Evidence:** \$FileCreateMode is set to 0640 in /etc/rsyslog.conf.
- **Recommendation:** None.

Rule 6.1.2.4: Ensure rsyslog logging is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** Rsyslog is configured with default logging rules in /etc/rsyslog.d/50-default.conf for auth, syslog, kern, mail, etc.
- **Recommendation:** Manually review rsyslog configuration files to ensure all necessary system events are logged according to organizational requirements.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log host (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** No remote log host configuration (e.g., *.* @@remote-log-host) is found in the provided rsyslog configuration files.
- **Recommendation:** Configure rsyslog to send logs to a remote, hardened log host to ensure log availability and integrity, as per organizational policy.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client (Automated)

- **Status:** PASS
- **Evidence:** imudp and imtcp modules for receiving remote logs are commented out in /etc/rsyslog.conf.
- **Recommendation:** None.

Rule 6.1.2.7: Ensure logrotate is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** Logrotate is installed, enabled, active, and configured with multiple rotation rules in /etc/logrotate.conf and /etc/logrotate.d/.
- **Recommendation:** Manually review logrotate configurations to ensure all critical log files are rotated, compressed, and retained according to organizational policy.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed (Automated)

- **Status:** FAIL
- **Evidence:** rsyslog-gnutls package is not installed.
- **Recommendation:** Install the rsyslog-gnutls package using `sudo apt install`

`rsyslog-gnutls` to enable secure remote log forwarding.

Rule 6.1.2.9: Ensure rsyslog forwarding uses gtls (Automated)

- **Status:** FAIL
- **Evidence:** `rsyslog-gnutls` is not installed, therefore secure forwarding using GTLS cannot be configured.
- **Recommendation:** Install `rsyslog-gnutls` and configure `rsyslog` to use GTLS for secure remote log forwarding.

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The security profile does not contain information about `rsyslog` CA certificate configuration.
- **Recommendation:** If remote log forwarding is configured, ensure that CA certificates are properly configured for `rsyslog` to establish trusted connections with remote log hosts.

Rule 6.1.3.1: Ensure access to all logfiles has been configured (Automated)

- **Status:** FAIL
- **Evidence:** Multiple log files have insecure permissions:
 - `/var/log/wtmp (0664)`, `/var/log/lastlog (0664)`, `/var/log/faillog (0644)`, `/var/log/apt/history.log (0644)`, `/var/log/landscape/sysinfo.log (0644)`, `/var/log/alternatives.log (0644)`, `/var/log/dpkg.log (0644)`, `/var/log/unattended-upgrades/unattended-upgrades.log (0644)`, `/var/log/unattended-upgrades/unattended-upgrades-shutdown.log (0644)` are world-readable.
 - `/var/log/btmp (0660)` is group-writable by `utmp`.
- **Recommendation:** Adjust permissions for all log files to `0640` or `0600` (owner `root`, group `adm` or `syslog`) to prevent unauthorized access. For example, `sudo chmod 0640 /var/log/wtmp`.

Section 6.2 – System Auditing

Rule 6.2.1.1: Ensure auditd packages are installed (Automated)

- **Status:** FAIL
- **Evidence:** `auditd` package is not installed.
- **Recommendation:** Install the `auditd` package using `sudo apt install auditd audispd-plugins`.

Rule 6.2.1.2: Ensure auditd service is enabled and active (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of its service status.
- **Recommendation:** Install the `auditd` package and ensure the service is enabled and active.

Rule 6.2.1.3: Ensure auditing for processes that start prior to auditd is enabled (Automated)

- **Status:** UNKNOWN
- **Evidence:** auditd package is not installed, preventing evaluation of pre-auditd process auditing.
- **Recommendation:** Install the auditd package and configure it to audit processes that start prior to the audit daemon.

Rule 6.2.1.4: Ensure audit_backlog_limit is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** auditd package is not installed, preventing evaluation of audit_backlog_limit.
- **Recommendation:** Install the auditd package and configure audit_backlog_limit in /etc/audit/audit.conf.

Rule 6.2.2.1: Ensure audit log storage size is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** auditd package is not installed, preventing evaluation of audit log storage size.
- **Recommendation:** Install the auditd package and configure audit log storage size in /etc/audit/auditd.conf.

Rule 6.2.2.2: Ensure audit logs are not automatically deleted (Automated)

- **Status:** UNKNOWN
- **Evidence:** auditd package is not installed, preventing evaluation of audit log deletion settings.
- **Recommendation:** Install the auditd package and configure auditd to prevent automatic deletion of logs.

Rule 6.2.2.3: Ensure system is disabled when audit logs are full (Automated)

- **Status:** UNKNOWN
- **Evidence:** auditd package is not installed, preventing evaluation of system behavior when logs are full.
- **Recommendation:** Install the auditd package and configure auditd to disable the system when audit logs are full.

Rule 6.2.2.4: Ensure system warns when audit logs are low on space (Automated)

- **Status:** UNKNOWN
- **Evidence:** auditd package is not installed, preventing evaluation of audit log space warning settings.
- **Recommendation:** Install the auditd package and configure auditd to warn when audit logs are low on space.

Rule 6.2.3.1: Ensure changes to system administration scope (sudoers) is collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect changes to `sudoers` files.

Rule 6.2.3.2: Ensure actions as another user are always logged (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to log actions as another user.

Rule 6.2.3.3: Ensure events that modify the sudo log file are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify the `sudo` log file.

Rule 6.2.3.4: Ensure events that modify date and time information are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify date and time information.

Rule 6.2.3.5: Ensure events that modify `sethostname` and `setdomainname` are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `sethostname` and `setdomainname`.

Rule 6.2.3.6: Ensure events that modify `/etc/issue` and `/etc/issue.net` are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/issue` and `/etc/issue.net`.

Rule 6.2.3.7: Ensure events that modify `/etc/hosts` and `/etc/hostname` are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/hosts` and `/etc/hostname`.

Rule 6.2.3.8: Ensure events that modify `/etc/network` and `/etc/networks` are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/network` and `/etc/networks`.

Rule 6.2.3.9: Ensure events that modify `/etc/netplan` are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/netplan`.

Rule 6.2.3.10: Ensure use of privileged commands are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect the use of privileged commands.

Rule 6.2.3.11: Ensure events that modify `/etc/group` information are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/group` information.

Rule 6.2.3.12: Ensure events that modify `/etc/passwd` information are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/passwd` information.

Rule 6.2.3.13: Ensure events that modify `/etc/shadow` and `/etc/gshadow` are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect

events that modify `/etc/shadow` and `/etc/gshadow`.

Rule 6.2.3.14: Ensure events that modify `/etc/security/opasswd` are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/security/opasswd`.

Rule 6.2.3.15: Ensure events that modify `/etc/nsswitch.conf` file are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/nsswitch.conf`.

Rule 6.2.3.16: Ensure events that modify `/etc/pam.conf` and `/etc/pam.d/` information are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify `/etc/pam.conf` and `/etc/pam.d/` information.

Rule 6.2.3.17: Ensure unsuccessful file access attempts are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect unsuccessful file access attempts.

Rule 6.2.3.18: Ensure discretionary access control permission modification events are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect discretionary access control permission modification events.

Rule 6.2.3.19: Ensure successful file system mounts are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect successful file system mounts.

Rule 6.2.3.20: Ensure session initiation information is collected

(Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect session initiation information.

Rule 6.2.3.21: Ensure login and logout events are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect login and logout events.

Rule 6.2.3.22: Ensure file deletion events by users are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect file deletion events by users.

Rule 6.2.3.23: Ensure events that modify the system's Mandatory Access Controls are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect events that modify the system's Mandatory Access Controls.

Rule 6.2.3.24: Ensure successful and unsuccessful attempts to use the `chcon` command are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect attempts to use the `chcon` command.

Rule 6.2.3.25: Ensure successful and unsuccessful attempts to use the `setfacl` command are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect attempts to use the `setfacl` command.

Rule 6.2.3.26: Ensure successful and unsuccessful attempts to use the `chacl` command are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.

- **Recommendation:** Install the `auditd` package and configure audit rules to collect attempts to use the `chacl` command.

Rule 6.2.3.27: Ensure successful and unsuccessful attempts to use the usermod command are collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect attempts to use the `usermod` command.

Rule 6.2.3.28: Ensure kernel module loading unloading and modification is collected (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit rules.
- **Recommendation:** Install the `auditd` package and configure audit rules to collect kernel module loading, unloading, and modification events.

Rule 6.2.3.29: Ensure the audit configuration is immutable (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit configuration immutability.
- **Recommendation:** Install the `auditd` package and configure the audit system to be immutable.

Rule 6.2.3.30: Ensure the running and on disk configuration is the same (Manual)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit configuration consistency.
- **Recommendation:** Install the `auditd` package. Once installed, manually verify that the running audit configuration matches the on-disk configuration.

Rule 6.2.4.1: Ensure the audit log file directory mode is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit log directory permissions.
- **Recommendation:** Install the `auditd` package and configure the audit log file directory mode to `0750`.

Rule 6.2.4.2: Ensure audit log files mode is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit log file permissions.
- **Recommendation:** Install the `auditd` package and configure the audit log files mode

to 0640.

Rule 6.2.4.3: Ensure audit log files owner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit log file owner.
- **Recommendation:** Install the `auditd` package and configure the audit log files owner to `root`.

Rule 6.2.4.4: Ensure audit log files group owner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit log file group owner.
- **Recommendation:** Install the `auditd` package and configure the audit log files group owner to `adm` or `syslog`.

Rule 6.2.4.5: Ensure audit configuration files mode is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit configuration file permissions.
- **Recommendation:** Install the `auditd` package and configure the audit configuration files mode to 0640.

Rule 6.2.4.6: Ensure audit configuration files owner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit configuration file owner.
- **Recommendation:** Install the `auditd` package and configure the audit configuration files owner to `root`.

Rule 6.2.4.7: Ensure audit configuration files group owner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit configuration file group owner.
- **Recommendation:** Install the `auditd` package and configure the audit configuration files group owner to `root`.

Rule 6.2.4.8: Ensure audit tools mode is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit tools permissions.

- **Recommendation:** Install the `auditd` package and configure the audit tools mode to 0755.

Rule 6.2.4.9: Ensure audit tools owner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit tools owner.
- **Recommendation:** Install the `auditd` package and configure the audit tools owner to `root`.

Rule 6.2.4.10: Ensure audit tools group owner is configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** `auditd` package is not installed, preventing evaluation of audit tools group owner.
- **Recommendation:** Install the `auditd` package and configure the audit tools group owner to `root`.

Section 6.3 – Configure Integrity Checking

Rule 6.3.1: Ensure AIDE is installed (Automated)

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.aide_installed` is false.
- **Recommendation:** Install the AIDE package using `sudo apt install aide aide-common`.

Rule 6.3.2: Ensure filesystem integrity is regularly checked (Automated)

- **Status:** UNKNOWN
- **Evidence:** AIDE is not installed, and no information about scheduled integrity checks is available.
- **Recommendation:** Install AIDE and configure a cron job or systemd timer to regularly check filesystem integrity (e.g., daily or weekly).

Rule 6.3.3: Ensure cryptographic mechanisms are used to protect the integrity of audit tools (Automated)

- **Status:** UNKNOWN
- **Evidence:** AIDE is not installed, and no information about cryptographic integrity protection for audit tools is available.
- **Recommendation:** Install AIDE and configure it to monitor the integrity of critical audit tools using strong cryptographic hashes (e.g., SHA512).

Section 7.1 – Configure system file and directory access

Rule 7.1.1: Ensure access to `/etc/passwd` is configured (Automated)

- **Status:** PASS
- **Evidence:** `/etc/passwd` has permissions 0644, owner `root`, group `root`.
- **Recommendation:** None.

Rule 7.1.2: Ensure access to /etc/passwd- is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/passwd- has permissions 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.3: Ensure access to /etc/group is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/group has permissions 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.4: Ensure access to /etc/group- is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/group- has permissions 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.5: Ensure access to /etc/shadow is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/shadow has permissions 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.6: Ensure access to /etc/shadow- is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/shadow- has permissions 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.7: Ensure access to /etc/gshadow is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/gshadow has permissions 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.8: Ensure access to /etc/gshadow- is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/gshadow- has permissions 0640, owner root, group shadow.
- **Recommendation:** None.

Rule 7.1.9: Ensure access to /etc/shells is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/shells has permissions 0644, owner root, group root.
- **Recommendation:** None.

Rule 7.1.10: Ensure access to /etc/security/opasswd is configured (Automated)

- **Status:** PASS
- **Evidence:** `/etc/security/opasswd` has permissions `0600`, owner `root`, group `root`.
- **Recommendation:** None.

Rule 7.1.11: Ensure world writable files and directories are secured (Automated)

- **Status:** PASS
- **Evidence:** No world-writable files or directories without the sticky bit are found on scanned mounts (`/`, `/boot`).
- **Recommendation:** None.

Rule 7.1.12: Ensure no files or directories without an owner and a group exist (Automated)

- **Status:** PASS
- **Evidence:** No unowned or ungrouped files or directories are found on scanned mounts (`/`, `/boot`).
- **Recommendation:** None.

Rule 7.1.13: Ensure SUID and SGID files are reviewed (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The following SUID/SGID files were found: `/usr/bin/umount`, `/usr/bin/chsh`, `/usr/bin/sudo`, `/usr/bin/su`, `/usr/bin/passwd`, `/usr/bin/ssh-agent`, `/usr/bin/fusermount3`, `/usr/bin/newgrp`, `/usr/bin/chage`, `/usr/bin/expiry`, `/usr/bin/gpasswd`, `/usr/bin/crontab`, `/usr/bin/mount`, `/usr/bin/chfn`, `/usr/lib/dbus-1.0/dbus-daemon-launch-helper`, `/usr/lib/x86_64-linux-gnu/utempter/utempter`, `/usr/lib/openssh/ssh-keysign`, `/usr/lib/polkit-1/polkit-agent-helper-1`, `/usr/sbin/pam_extrausers_chkpwd`, `/usr/sbin/unix_chkpwd`.
- **Recommendation:** Manually review the list of SUID/SGID files to ensure that only authorized and necessary programs have these permissions. Remove SUID/SGID bits from any files that do not require them.

Section 7.2 – Local User and Group Settings

Rule 7.2.1: Ensure accounts in `/etc/passwd` use shadowed passwords (Automated)

- **Status:** PASS
- **Evidence:** All user accounts listed in `user_chage_info` have a `passwd_status` of `*` or `L`, indicating shadowed passwords are in use.
- **Recommendation:** None.

Rule 7.2.2: Ensure `/etc/shadow` password fields are not empty (Automated)

- **Status:** PASS
- **Evidence:** All user accounts in `/etc/shadow` have non-empty password fields

(indicated by * or L status).

- **Recommendation:** None.

Rule 7.2.3: Ensure all groups in /etc/passwd exist in /etc/group (Automated)

- **Status:** PASS
- **Evidence:** No orphan GIDs (groups in /etc/passwd not found in /etc/group) were detected.
- **Recommendation:** None.

Rule 7.2.4: Ensure shadow group is empty (Automated)

- **Status:** PASS
- **Evidence:** The shadow group has no members.
- **Recommendation:** None.

Rule 7.2.5: Ensure no duplicate UIDs exist (Automated)

- **Status:** PASS
- **Evidence:** No duplicate UIDs were detected.
- **Recommendation:** None.

Rule 7.2.6: Ensure no duplicate GIDs exist (Automated)

- **Status:** PASS
- **Evidence:** No duplicate GIDs were detected.
- **Recommendation:** None.

Rule 7.2.7: Ensure no duplicate user names exist (Automated)

- **Status:** PASS
- **Evidence:** No duplicate user names were detected.
- **Recommendation:** None.

Rule 7.2.8: Ensure no duplicate group names exist (Automated)

- **Status:** PASS
- **Evidence:** No duplicate group names were detected.
- **Recommendation:** None.

Rule 7.2.9: Ensure local interactive user home directories are configured (Automated)

- **Status:** PASS
- **Evidence:** No issues detected with local interactive user home directories (e.g., missing, incorrect permissions).
- **Recommendation:** None.

Rule 7.2.10: Ensure local interactive user dot files access is configured (Automated)

- **Status:** PASS
 - **Evidence:** No issues detected with local interactive user dot file permissions.
 - **Recommendation:** None.
-

SUID/SGID Triage Report

```
{ "findings": [], "summary": { "suspicious_found": 0, "message": "No unexpected SUID/SGID files detected." } }
```